Appendix G
Physics of Sports

Many of the motions important in sports are analyzed in Chaps. 3 and 7 using simple physics. This includes the actual motion of the body when throwing a ball or hitting it with a bat or racket, the flight of the ball, and collisions. Other chapters address additional aspects of sports activities, such as injuries and helmets (in Chap. 4), the action of muscles (Chap. 5), increased metabolic rates and need for body cooling (Chap. 6), and increased breathing rates (Chap. 9). As such, this appendix addresses important aspects of physics by the body. Moreover, we have more specifically examined the physics of several sports, including those in track and field, baseball, and boxing.

There are several books devoted to the physics of sports; some focus on one sport and others examine several sports. Some emphasize the object of the sport and not how the human body generates the needed action. (They are nonetheless very interesting.) The excellent book by Hay [17] is one of several exceptions in that it examines the biomechanics of many sports, including gymnastics, swimming, track and field, and several major team sports. Equally excellent is the insightful overview by Brancazio [4], which describes the physics of the many features of sports activities simply, yet accurately and comprehensively. The overview by Dyson [10] emphasizes areas not always covered by others, such as gymnastics and track and field events. Other nice overviews of many sports can be found in [27].

An excellent compendium of articles describing many sports can be found in the book by Armenti [3], including those on archery, badminton, baseball, basketball, bowling, canoeing, cycling, discus throwing, diving, football, golf, jumping, karate, rowing, running, sailing, shot putting, skiing, swimming, and tennis. Most focus on the motions themselves, but some focus on metabolism. You can get a good feel for the range of the physics in athletics by perusing through these articles or the cited books by Brancazio, Dyson and Hay. Comprehensive bibliographies of articles and other resources on the physics of sports can be found in [13, 14].

The book edited by Hung and Pallis [18] examines the physics of golf, tennis, baseball, football and soccer, and basketball. Adair [1, 2] and Watts and Bahill [28] have analyzed baseball, concentrating on the motion of the ball during a pitch and after it has been hit. Other valuable books and articles about baseball include [7, 25]. Gay [15] discusses the physics of football. The physics of basketball is covered in

© Springer International Publishing Switzerland 2016

I.P. Herman, *Physics of the Human Body*, Biological and Medical Physics,
Biomedical Engineering, DOI 10.1007/978-3-319-23932-3

[12] and of tennis in [5]. Several aspects of cycling are covered in [6]; Brancazio [4] has a nice discussion of the stability of bicycles. Jorgensen [19] has examined golf. Golf is also highlighted in the book by [8]. Lind and Sanders [14] have looked at the physics of skiing, including the skiing motion, snow, ski wax, and so on.

The flights of balls and other projectiles are covered in [4, 7, 8, 16, 26]. In particular, the books by Daish [8] and de Mestre [9] solve the physics at a high level. The former book is a bit more comprehensive, while the latter tackles some more difficult problems.

The books by Laws and co-workers [11, 20–23] examine a subject related to the physics of sports: the physics of dancing.

The text discussions of the following sports, motions, and sports equipment can be located from the following listings in the index: Archery and arrows; Balls; Baseball; Basketball; Batting; Bicycle; Boating; Bouncing; Bowling; Boxing; Carrying; Catching; Climbing, ascending, descending stairs; Cricket; Cycling; Dancing; Diving; Exercise; Football; Flying, human flight; Golf; Gymnastics; Hammer throwing; High jump; Hockey; Hopping; Jumping; Karate; Kicking; Lifting; Long jump; Marathon running; Ping pong, table tennis; Pitching; Pole vault; Pool, billiards; Pulling, pushing; Pushups; Race walking; Rolling ball; Running; Shoveling; Shot putting; Sitting; Skating; Skidding ball; Skiing; Skipping; Sliding; Slipping; Soccer; Somersaulting; Spinning ball; Sports injuries; Sprinting; Stairs; Standing; Striking, hitting, punching; Swimming; Swinging; Tennis; Throwing; Treadmill; Vertical jump; Walking; Weightlifting, Training; and Wrestling.

Check "Sports injuries" in the index for injuries resulting from athletic events.

References

1. R.K. Adair, *The Physics of Baseball*, 3rd edn. (HarperCollins, New York, 2002)
2. R.K. Adair, The physics of baseball. Phys. Today 26–31 (1995)
3. A. Armenti Jr. (ed.), *The Physics of Sports, American Institute of Physics* (Springer, New York, 1992)
4. P.J. Brancazio, *Sport Science: Physical Laws and Optimum Performance* (Touchstone, New York, 1984)
5. H. Brody, R. Cross, C. Lindsey, *The Physics and Technology of Tennis*, (Usrsa, Vista, California, 2004)
6. E.R. Burke (ed.), *Science of Cycling* (Human Kinetics, Champaign, Illinois, 1986)
7. R. Cross, *Physics of Baseball and Softball*, (Springer, New York, 2011)
8. C.B. Daish, *The Physics of Ball Games* (The English Universities Press, London, 1972)
9. N. de Mestre, *The Mathematics of Projectiles in Sport* (Cambridge University Press, Cambridge, 1990)
10. G.H.G. Dyson, B.D. Woods, P.R. Travers, *Dyson's Mechanics of Athletics*, 8th edn (Holmes & Meier, New York, 1986)
11. S.S. Fitt, *Dance Kinesiology*, 2nd edn. (Schirmer/Thomson Learning, New York, 1996)
12. J.J. Fontanella, *The Physics of Basketball* (The Johns Hopkins University Press, Baltimore, 2006)
13. C. Frohlich, Resource letter PS-1: physics of sports. Am. J. Phys. **54**, 590–593 (1986)
14. C. Frohlich, Resource letter PS-2: physics of sports. Am. J. Phys. **79**, 565–574 (2011)
15. T. Gay, *The Physics of Football: Discover the Science of Bone-Crunching Hits, Soaring Field Goals, and Awe-Inspiring Passes* (HarperCollins, New York, 2005)

16. D.F. Griffing, *The Dynamics of Sports: Why That's the Way the Ball Bounces*, 4th edn. (Dalog, Osford, Ohio, 1995)
17. J.G. Hay, *The Biomechanics of Sports Techniques*, 4th edn. (Prentice-Hall, Englewood Cliffs, 1993)
18. G.K. Hung, J.M. Pallis (eds.), *Biomedical Engineering Principles in Sports* (Kluwer/Plenum, New York, 2004)
19. T.P. Jorgensen, *The Physics of Golf*, 2nd edn. (AIP/Springer, New York, 1999)
20. K. Laws, *The Physics of Dance*, (Schirmer Books, New York, 1984)
21. K. Laws, C. Harvey, M. Swope, *Physics, Dance, and the Pas De Deux*, (Schirmer Books, New York, 1994)
22. K. Laws, M. Swope, *Physics and the Art of Dance: Understanding Movement* (Oxford University Press, Oxford, 2002)
23. K. Laws, M. Lott, Resource letter PoD-1: physics of dance. Am. J. Phys. **81**, 7–13 (2013)
24. D. Lind, S.P. Sanders, *The Physics of Skiing: Skiing at the Triple Point* (AIP/Springer, New York, 1996)
25. A.M. Nathan, L.V. Smith, W.L. Faber, D.A. Russell, Corked bats, juiced balls, and humidors: the physics of cheating in baseball. Am. J. Phys. **79**, 575–580 (2011)
26. M.S. Townend, *Mathematics in Sport* (Ellis Horwood Ltd, West Sussex, 1984)
27. F. Vizard (ed.), *Why a Curveball Curves: The Incredible Science of Sports (Popular Mechanics)* (Hearst , New York, 2009)
28. R.G. Watts, A.T. Bahill, *Keep Your Eye on the Ball: Curve Balls, Knuckleballs, and Fallacies of Baseball* (Freeman, New York, 2000)